

Azure Platform Case Study

Development / Testing / Production

A private application platform with Terraform and federated CI/CD

Three environments. Two subscriptions. Separate workload identities and private networks. Production approval, PIM and runtime verification.

Peter Hudcovic Evidence and build records reviewed October 6, 2026

Overview and subscriptions

Configuration and observed results

I built a private Azure application platform in Sweden Central with separate development, testing and production environments. Dev and Test share the non-production subscription. Production has its own subscription, identities, network and Terraform state. The shared package store lives in the Dev foundation resource group.

Home > Subscriptions

Subscriptions

Compare the different Azure subscription types

Forecast my cost for next billing period

+ Add

Manage Policies

View Requests

View eligible subscriptions

Export to CSV

Analyze

Global administrators can manage all subscriptions in this list by updating their policy setting [here](#).

Showing subscriptions in Default Directory directory. Don't see a subscription? [Switch directories](#)

Search for any field...

Subscriptions : Filtered (2 of 2)

My role == all

Status == all

Add filter

☐ Subscription name ↑↓	Subscription ID ↑↓	My role ↑↓	Current cost	Secure Score ↑↓	Parent management group ↑↓	Status ↑↓
☐ sub-app-nonprod	8b38e1d5-139f-4006-8f2b-ffb32a6ac114	Owner	€2.39	-	Tenant Root Group	Active
☐ sub-app-prod	61845030-2daa-4797-a97a-59a7fd8d478f	Owner	0.00	-	Tenant Root Group	Active

Environment comparison

Actual scope and role capabilities

	Development	Testing	Production
Subscription	sub-app-nonprod	sub-app-nonprod	sub-app-prod
VNet CIDR	10.10.0.0/16	10.20.0.0/16	10.30.0.0/16
Gateway private IP	10.10.4.10	10.20.4.10	10.30.4.10
App Service plan	B1	P0v3	P0v3
Monitoring	Dev workspace	Shared Dev workspace	Own workspace
Bastion SKU	Developer	Developer	Basic
Application delivery	main completion trigger	main completion trigger	Manual approval
Privileged access	Lab bootstrap admin	Lab bootstrap admin	Reader + VM login, PIM roles

Subscriptions and shared delivery

Architecture derived from the source snapshot



No VNet peering. Private DNS links remain environment-specific.

Users and access

Configuration and observed results

Application sign-in and infrastructure administration use separate grants. Lab Admin bootstraps the lab and belongs to the app-user groups. Prod Admin has permanent Reader and Virtual Machine User Login, with privileged resource and secret roles available through PIM. PIM activation requires justification and MFA and lasts at most two hours. Conditional Access policies evaluate non-production NAT locations and device-code sign-in in report-only mode.

The screenshot shows the 'Access control (IAM)' page for the 'sub-app-prod' subscription. The 'Role assignments' tab is selected, displaying a search for 'prod admin' which returned 2 results. The results are categorized into 'Contributor (1)' and 'Reader (1)'. The table below shows the details of these assignments.

Contributor (1)	Role	Scope	State	End time	Condition
☐ Contributor (1)					
☐ Prod Admin 0ba401df-650b-4548-9360-bf5...	Contributor	This resource	Eligible Permanent	Permanent	None
☒ Reader (1)					
☒ Prod Admin 0ba401df-650b-4548-9360-bf5...	Reader	This resource	Active Permanent	Permanent	None

Showing 1 - 2 of 2 results.

Source: [infra/environments/production/layer-0-foundation/access.tf](https://github.com/peterhudcovic/azure-platform-lab/blob/main/infra/environments/production/layer-0-foundation/access.tf)

Access and permissions matrix

Actual scope and role capabilities

Account / identity	Scope and capability	Boundary
Lab Admin	Bootstrap across all environments; app-group member	Broad administrator, no declared Entra VM login grant
Prod Admin	Reader + VM User Login permanently; PIM Contributor and secret roles	Prod scope; privileged roles need activation
App-user groups	Sign-in to assigned environment application	No Azure management role; /health excluded
Application MI	Secrets User on own application-vault RG	Secret read; no secret write grant
Gateway MI	Secrets User on certificate vault	Vault scope, not a single certificate
Build MI	Blob Data Contributor on shared packages	Read/write/delete blobs, not upload only
Deploy MI	Website Contributor on own app RG; package Blob Reader	Website management exceeds package deploy
Infra L1 / L2	Contributor on own layer RGs; state and dependency grants	No role assignment; resource changes can use runtime MIs
Destroy MI	Contributor on network/app/vault RGs and state access	Create/update/delete, not delete only
Prod lock roles	Infra L2 creates locks; destroy removes locks	Explicit custom lock grants on app/vault RGs
DevOpsInfrastructure	Reader + Network Contributor on network RG	Pool placement, not application data access

Managed identities and role boundaries

Configuration and observed results

Each environment has six user-assigned identities for the application, gateway, deploy, destroy and two infrastructure layers. Dev foundation also hosts the shared build identity. Runtime identities read secrets at their assigned scopes. Pipeline identities use workload identity federation. Contributor and Website Contributor are broader than a single operation, so I document the actual role scopes instead of treating them as strict deploy-only or delete-only permissions.

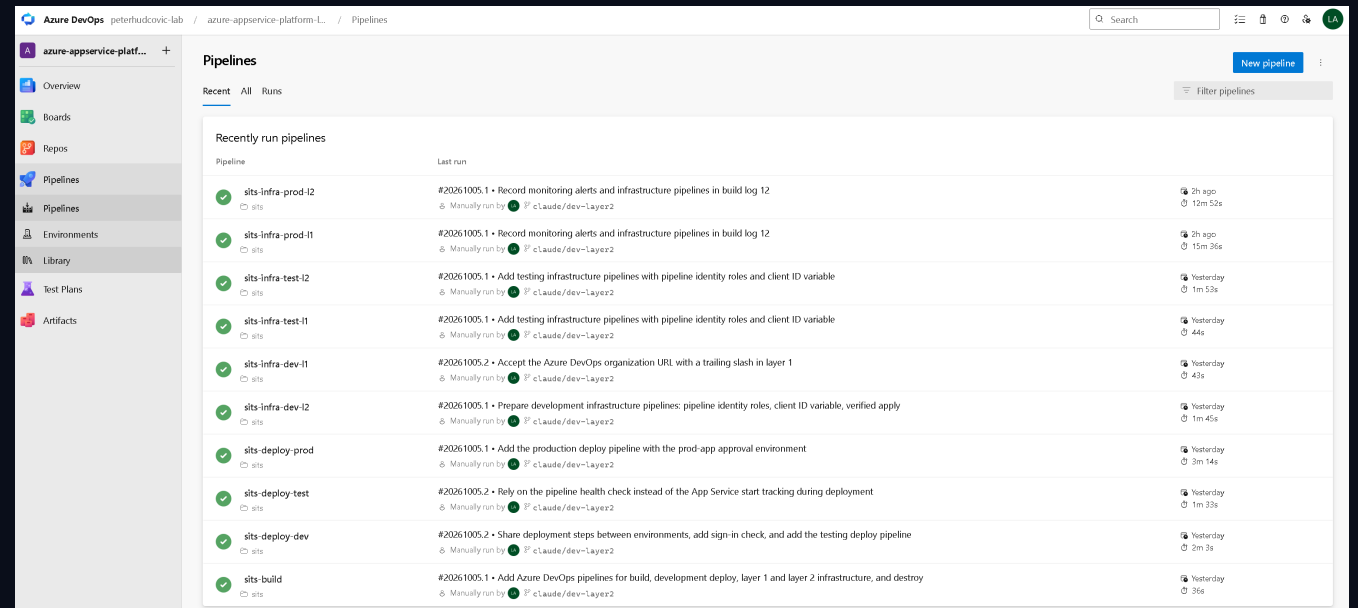
Managed Identities						
Default Directory (peterhudcovic@gmail.onmicrosoft.com)						
+ Create Manage view Refresh Export to CSV Open query Assign tags Delete Add to service group						
Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter						
☐	Name	Type	Resource Group	Location	Subscription ID ↑	
☐	 id-sits-prod-agw-swc	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f	
☐	 id-sits-prod-app-swc	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f	
☐	 id-sits-prod-deploy-swc	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f	
☐	 id-sits-prod-destroy-swc	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f	
☐	 id-sits-prod-infra-l1-swc	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f	
☐	 id-sits-prod-infra-l2-swc	Managed Identity	rg-sits-prod-foundation-swc	Sweden Central	/subscriptions/61845030-2daa-4797-a97a-59a7fd8d478f	
☐	 id-sits-build-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-dev-agw-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-dev-app-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-dev-deploy-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-dev-destroy-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-dev-infra-l1-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-dev-infra-l2-swc	Managed Identity	rg-sits-dev-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-test-agw-swc	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-test-app-swc	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-test-deploy-swc	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-test-destroy-swc	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-test-infra-l1-swc	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	
☐	 id-sits-test-infra-l2-swc	Managed Identity	rg-sits-test-foundation-swc	Sweden Central	/subscriptions/8b38e1d5-139f-4006-8f2b-ffb32a6ac114	

Source: docs/build-log/13-access-matrix.md

Shared CI/CD

Configuration and observed results

Azure DevOps builds one Python package, runs unit tests, records the source commit and SHA-256, and uploads the package to shared storage. Deploy jobs retrieve and verify the same package through environment-specific identities and private application agent pools. YAML enables Dev and Test completion triggers for main builds. Captured lab runs were manual on the working branch. Production deployment requires approval on prod-app, with self-approval enabled in this lab.



The screenshot shows the Azure DevOps web interface for a project named 'peterhudcovic-lab'. The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines, Environments, Library, Test Plans, and Artifacts. The main area is titled 'Pipelines' and shows a list of 'Recently run pipelines'. The table has columns for Pipeline, Last run, and a status icon. The pipelines listed are:

Pipeline	Last run	Status
sits-infra-prod-12	#20261005.1 • Record monitoring alerts and infrastructure pipelines in build log 12 Manually run by c.Laude/dev-Layer2	2h ago 12m 52s
sits-infra-prod-11	#20261005.1 • Record monitoring alerts and infrastructure pipelines in build log 12 Manually run by c.Laude/dev-Layer2	2h ago 15m 36s
sits-infra-test-12	#20261005.1 • Add testing infrastructure pipelines with pipeline identity roles and client ID variable Manually run by c.Laude/dev-Layer2	Yesterday 1m 53s
sits-infra-test-11	#20261005.1 • Add testing infrastructure pipelines with pipeline identity roles and client ID variable Manually run by c.Laude/dev-Layer2	Yesterday 44s
sits-infra-dev-11	#20261005.2 • Accept the Azure DevOps organization URL with a trailing slash in layer 1 Manually run by c.Laude/dev-Layer2	Yesterday 43s
sits-infra-dev-12	#20261005.1 • Prepare development infrastructure pipelines: pipeline identity roles, client ID variable, verified apply Manually run by c.Laude/dev-Layer2	Yesterday 1m 45s
sits-deploy-prod	#20261005.1 • Add the production deploy pipeline with the prod-app approval environment Manually run by c.Laude/dev-Layer2	Yesterday 3m 14s
sits-deploy-test	#20261005.2 • Rely on the pipeline health check instead of the App Service start tracking during deployment Manually run by c.Laude/dev-Layer2	Yesterday 1m 33s
sits-deploy-dev	#20261005.2 • Share deployment steps between environments, add sign-in check, and add the testing deploy pipeline Manually run by c.Laude/dev-Layer2	Yesterday 2m 3s
sits-build	#20261005.1 • Add Azure DevOps pipelines for build, development deploy, layer 1 and layer 2 infrastructure, and destroy Manually run by c.Laude/dev-Layer2	Yesterday 36s

Source: [pipelines/templates/deploy-steps.yml](#)

Terraform layers and infrastructure delivery

Configuration and observed results

Layer 0 retains foundation resources and identities. Layer 1 creates networking and operations. Layer 2 creates the application platform. Every environment and layer has a separate state container with Entra authentication. Layer 1 pipelines use hosted agents because they create the private pools. Layer 2 uses the environment infrastructure pool. Production has a plan stage and an approval on prod-infra, followed by a new plan, conditional apply and verification. Build log 12 records No changes runs in all environments.

peterhudcovic-lab / azure-appservice-platform-l... / Settings / Agent pools

Project Settings

azure-appservice-platform-lab

General

- Overview
- Teams
- Permissions
- Notifications
- Service hooks
- Dashboards

Boards

- Project configuration
- Team configuration
- GitHub connections

Pipelines

- Agent pools**
- Parallel jobs
- Settings
- Test management
- Service connections

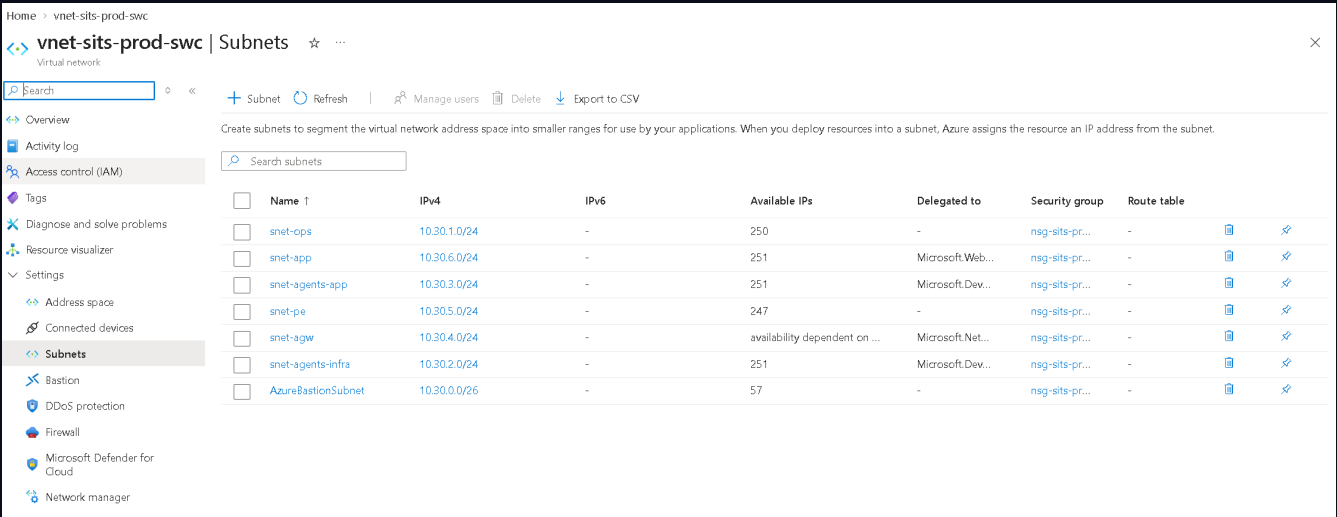
Agent pools

Name
 Azure Pipelines Azure Pipelines
 Default Azure Pipelines
 GitHub-hosted Agents Azure Pipelines
 mdp-sits-dev-app-swc Lab Admin
 mdp-sits-dev-infra-swc Lab Admin
 mdp-sits-prod-app-swc Lab Admin
 mdp-sits-prod-infra-swc Lab Admin
 mdp-sits-test-app-swc Lab Admin
 mdp-sits-test-infra-swc Lab Admin

Networking and private access

Configuration and observed results

The VNets use 10.10.0.0/16, 10.20.0.0/16 and 10.30.0.0/16 with no peering. Subnets separate operations, gateway, application integration, private endpoints and pipeline agents. Private DNS zones link to their own VNet. App Service inbound private endpoints and outbound VNet integration serve different directions of traffic. NSGs use explicit permitted flows followed by deny-all inbound at priority 4000. NAT attaches to operations, application and agent subnets.



The screenshot shows the 'Subnets' page in the Azure portal for the virtual network 'vnet-sits-prod-swc'. The page includes a left-hand navigation pane with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Connected devices, Subnets (selected), Bastion, DDoS protection, Firewall, Microsoft Defender for Cloud, and Network manager. The main content area displays a table of subnets with columns for Name, IPv4, IPv6, Available IPs, Delegated to, Security group, and Route table. There are also icons for deleting and editing each subnet.

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	snet-ops	10.30.1.0/24	-	250	-	nsg-sits-pr...	-		
☐	snet-app	10.30.6.0/24	-	251	Microsoft.Web...	nsg-sits-pr...	-		
☐	snet-agents-app	10.30.3.0/24	-	251	Microsoft.Dev...	nsg-sits-pr...	-		
☐	snet-pe	10.30.5.0/24	-	247	-	nsg-sits-pr...	-		
☐	snet-agw	10.30.4.0/24	-	availability dependent on ...	Microsoft.Net...	nsg-sits-pr...	-		
☐	snet-agents-infra	10.30.2.0/24	-	251	Microsoft.Dev...	nsg-sits-pr...	-		
☐	AzureBastionSubnet	10.30.0.0/26	-	57	-	nsg-sits-pr...	-		

Source: docs/build-log/14-acceptance-checks-and-certificate-trust.md

Production traffic paths

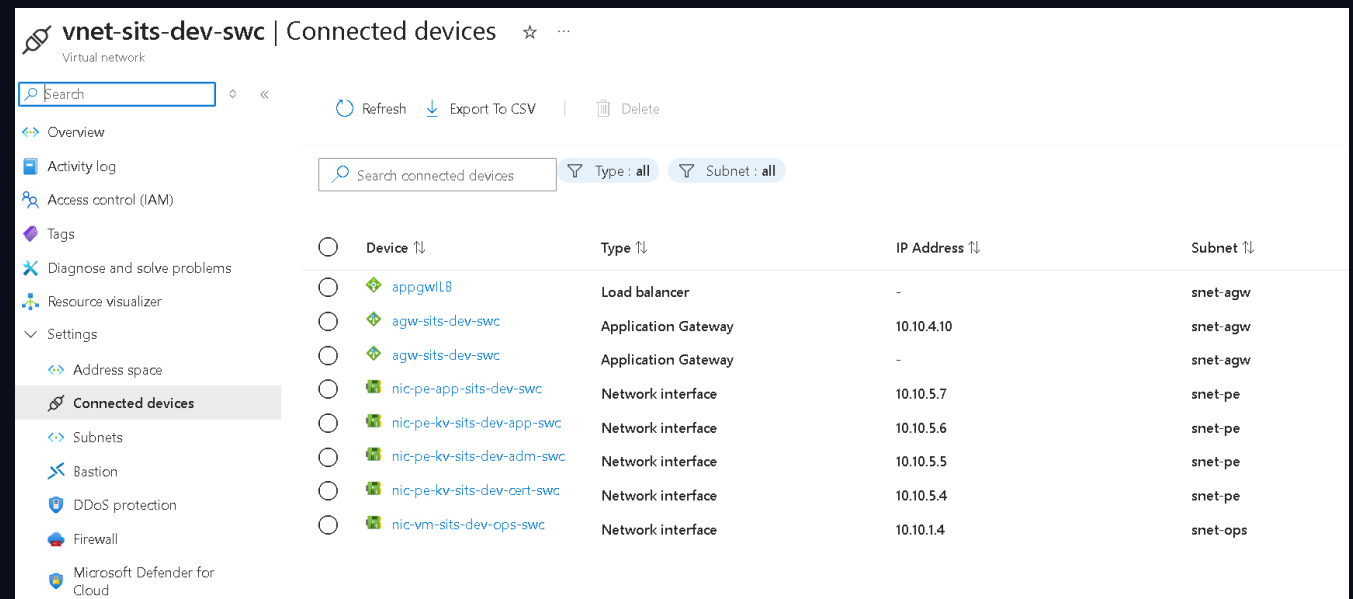
DNS resolves addresses separately from application HTTPS traffic

Flow	Origin	Destination and purpose
Operations session	Azure portal / Bastion	Ops VM; Prod Entra user login or local emergency account
Application HTTPS	Ops VM 10.30.1.4	Gateway private listener 10.30.4.10:443
Backend HTTPS	Application Gateway	App private endpoint 10.30.5.7:443, then App Service
Deployment	Private application pool	App SCM private endpoint; package fingerprint verified
Application secrets	App outbound VNet integration	Application Key Vault private endpoint 10.30.5.6:443
Listener certificate	Gateway managed identity	Certificate vault secret, separately authorized
DNS resolution	VNet clients	Internal and privatelink zones; DNS is not an HTTPS hop
Diagnostics	Gateway and application vault	Environment Log Analytics and e-mail action groups

Development

Configuration and observed results

Development uses sub-app-nonprod and vnet-sits-dev-swc with address space 10.10.0.0/16. The private gateway is 10.10.4.10 and the application hostname is app.dev.sits.internal. App Service uses B1. Development hosts the shared package store and the monitoring workspace used by both non-production environments. Separate application and infrastructure pools run jobs within the Dev network.

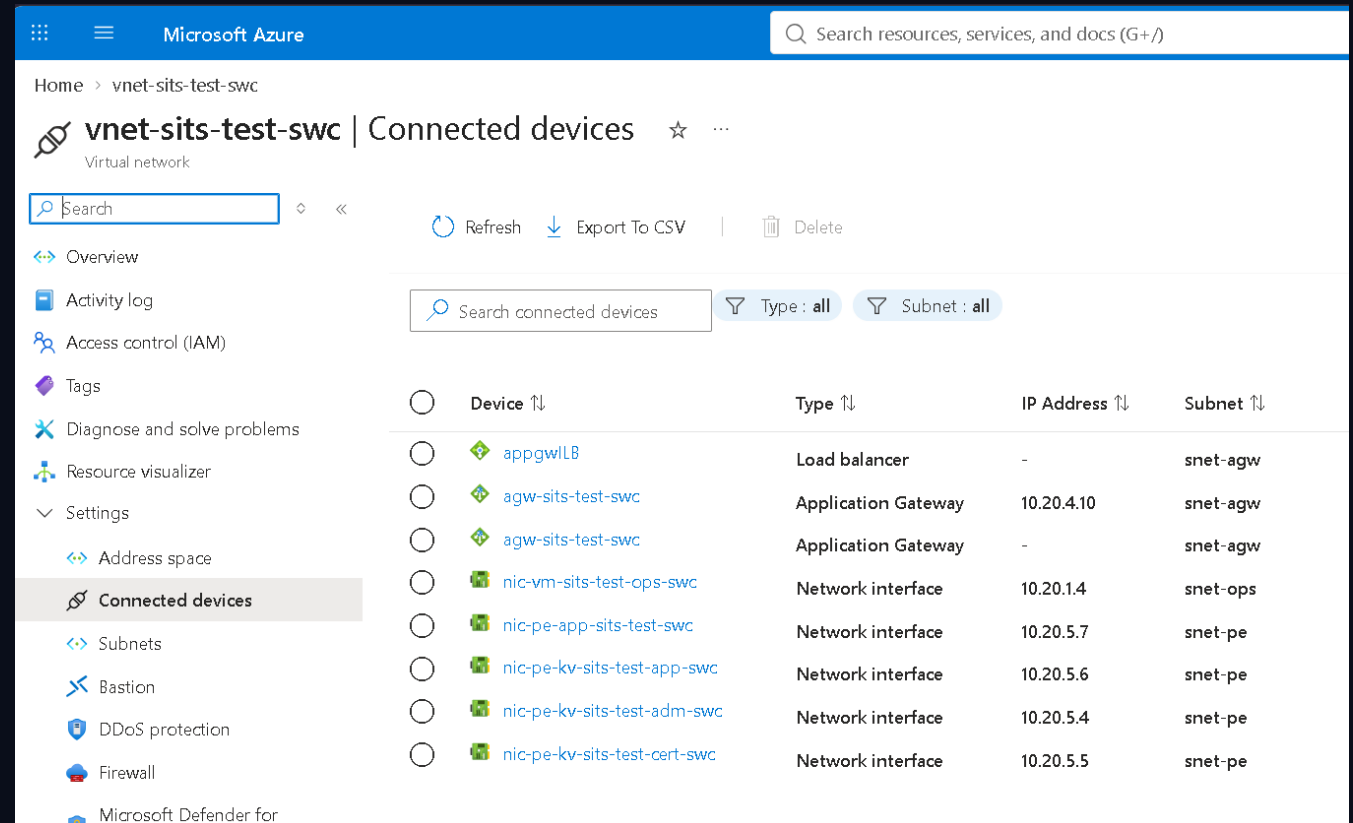


Device	Type	IP Address	Subnet
apgw-lb	Load balancer	-	snet-agw
agw-sits-dev-swc	Application Gateway	10.10.4.10	snet-agw
agw-sits-dev-swc	Application Gateway	-	snet-agw
nic-pe-app-sits-dev-swc	Network interface	10.10.5.7	snet-pe
nic-pe-kv-sits-dev-app-swc	Network interface	10.10.5.6	snet-pe
nic-pe-kv-sits-dev-adm-swc	Network interface	10.10.5.5	snet-pe
nic-pe-kv-sits-dev-cert-swc	Network interface	10.10.5.4	snet-pe
nic-vm-sits-dev-ops-swc	Network interface	10.10.1.4	snet-ops

Testing

Configuration and observed results

Testing uses the same non-production subscription but its own resource groups, VNet, identities and state. The VNet is 10.20.0.0/16, the gateway is 10.20.4.10 and the hostname is app.test.sits.internal. App Service uses P0v3. Testing reads the shared package and reuses the Dev Log Analytics and Application Insights resources. This is an explicit monitoring dependency rather than complete resource independence.



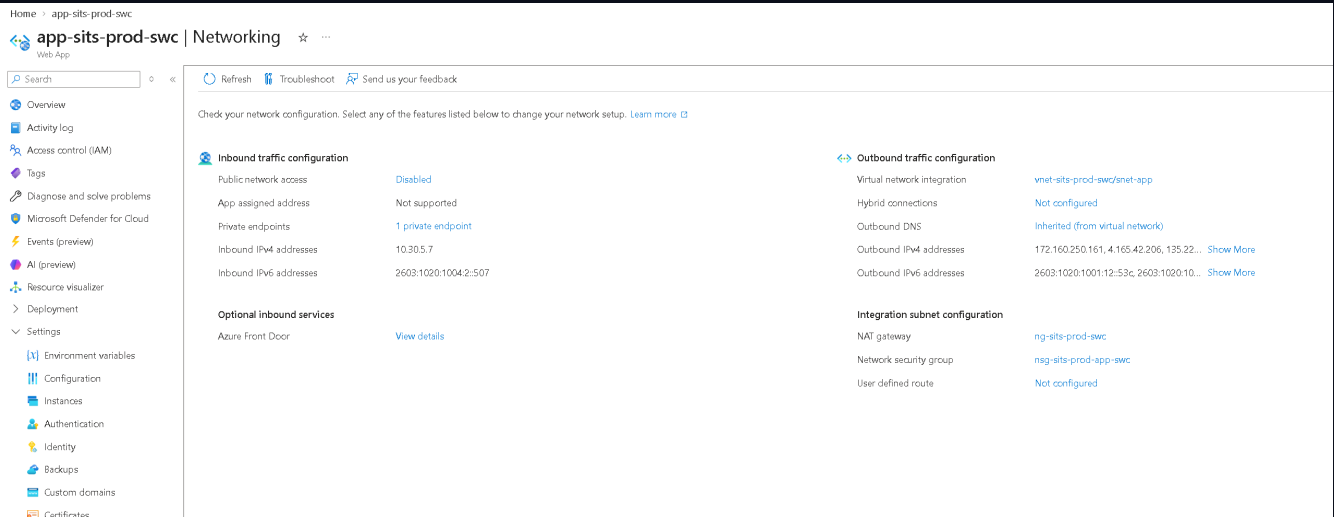
Device	Type	IP Address	Subnet
appgwLB	Load balancer	-	snet-agw
agw-sits-test-swc	Application Gateway	10.20.4.10	snet-agw
agw-sits-test-swc	Application Gateway	-	snet-agw
nic-vm-sits-test-ops-swc	Network interface	10.20.1.4	snet-ops
nic-pe-app-sits-test-swc	Network interface	10.20.5.7	snet-pe
nic-pe-kv-sits-test-app-swc	Network interface	10.20.5.6	snet-pe
nic-pe-kv-sits-test-adm-swc	Network interface	10.20.5.4	snet-pe
nic-pe-kv-sits-test-cert-swc	Network interface	10.20.5.5	snet-pe

Source: [docs/build-log/10-testing-environment.md](#)

Production

Configuration and observed results

Production uses sub-app-prod and 10.30.0.0/16. App Service uses P0v3 and the private gateway listens at 10.30.4.10 for app.prod.sits.internal. Production has its own monitoring, Bastion Basic, delete locks, PIM and separate approval environments. Public access is disabled on the Web App and application Key Vault. Certificate and admin foundation vaults retain public access, even though private endpoints also exist.

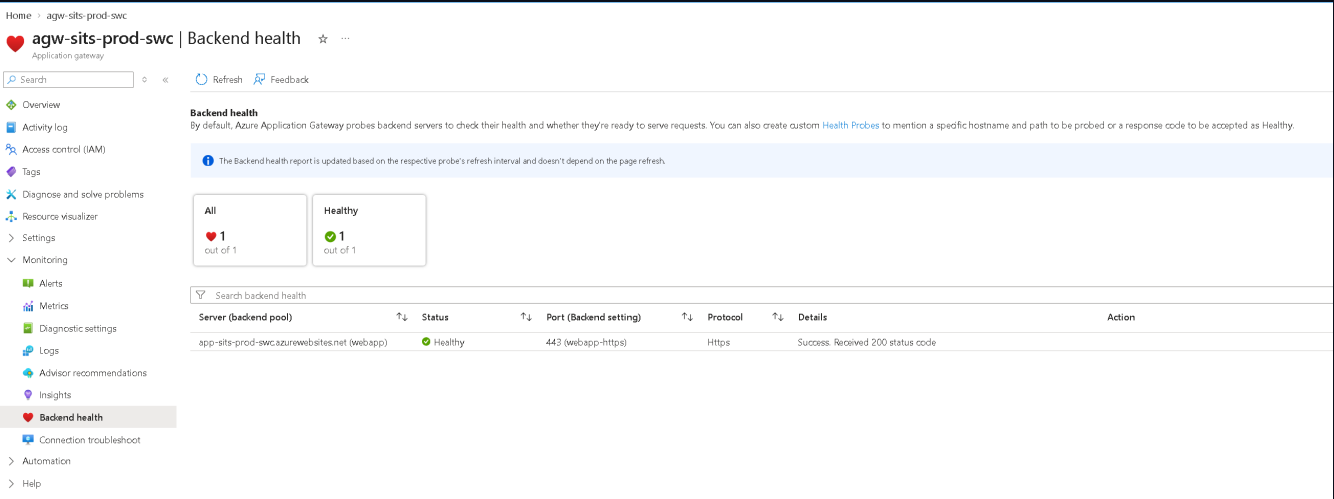


Source: docs/build-log/11-production-environment.md

Gateway, WAF and TLS

Configuration and observed results

The WAF_v2 gateway exposes a private HTTPS listener and forwards to the App Service hostname using HTTPS on port 443. A custom /health probe returned HTTP 200 and the backend was Healthy at capture time. Microsoft Default Rule Set 2.1 runs in Detection mode. A managed identity retrieves the listener certificate from the certificate vault. The self-signed certificate is trusted locally on each Ops VM as recorded in build log 14, and must be imported again after renewal or VM rebuild.

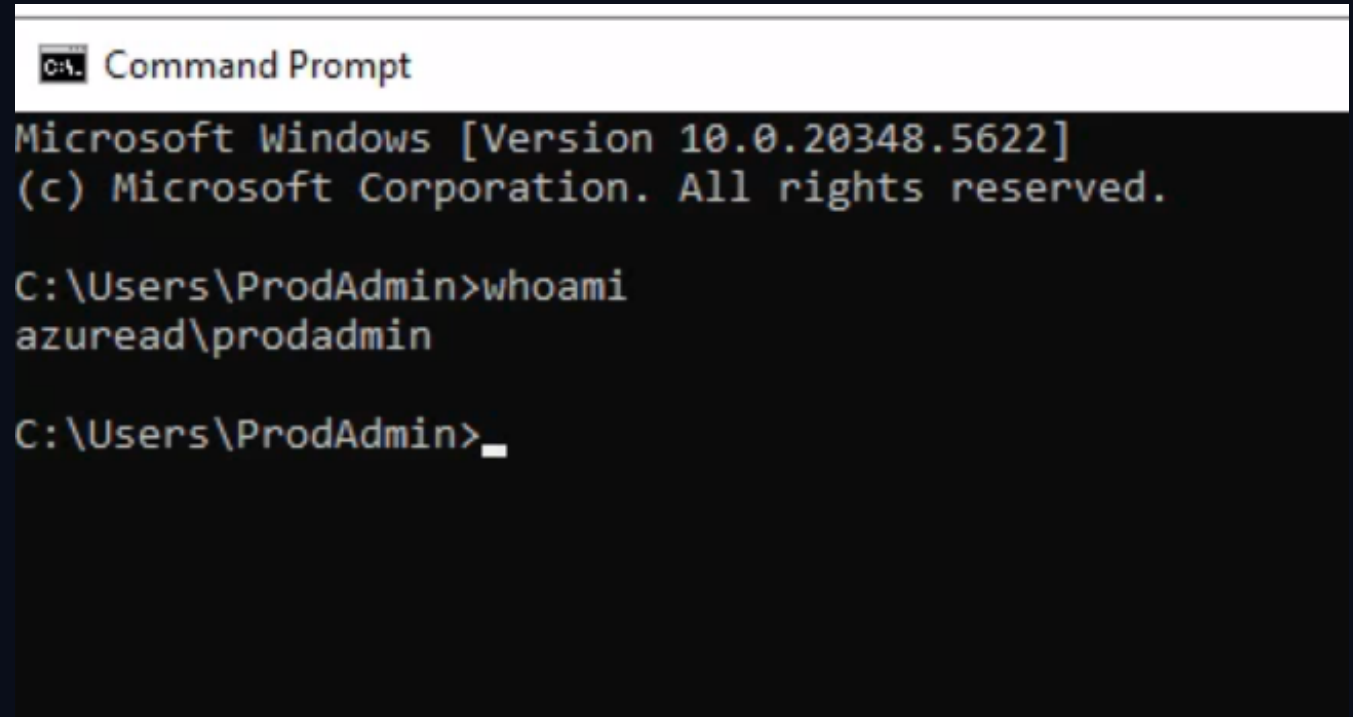


Source: docs/build-log/14-acceptance-checks-and-certificate-trust.md

Operations access through Bastion

Configuration and observed results

Bastion provides the operations entry point without a public IP on the VM NIC. Production supports Microsoft Entra ID sign-in through AADLoginForWindows and the permanent VM User Login grant. My captured session returned azuread\prodadmin. The local opsadmin account remains a separate emergency administrator path. The Entra user session does not need the local VM password or an activated Contributor role.



```
Command Prompt
Microsoft Windows [Version 10.0.20348.5622]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ProdAdmin>whoami
azuread\prodadmin

C:\Users\ProdAdmin>
```

Runtime validation from the production VM

Configuration and observed results

From source 10.30.1.4, app.prod.sits.internal resolved to the gateway at 10.30.4.10 and TCP port 443 was reachable. App Service resolved through privatelink.azurewebsites.net to 10.30.5.7. The application vault resolved through privatelink.vaultcore.azure.net to 10.30.5.6. These terminal outputs prove private DNS resolution and gateway TCP reachability. Build log 14 separately records trusted HTTPS /health responses and cross-environment DNS isolation tests.

```
Name                                     Type      TTL      Section      IPAddress
app.prod.sits.internal                   A         300      Answer       10.30.4.10
ComputerName                            : app.prod.sits.internal
RemoteAddress                           : 10.30.4.10
ResolvedAddresses                        : (10.30.4.10)
PingSucceeded                           : False
PingReplyDetails                         :
TcpClientSocket                         :
PingTestSucceeded                       : True
RemotePort                              : 443
TraceRoute                              :
Detailed                               : False
InterfaceAlias                          : Ethernet
InterfaceIndex                          : 6
InterfaceDescription                    : Microsoft Hyper-U Network Adapter
NetAdapter                              : MSFT_NetAdapter (CreationClassName = "MSFT_NetAdapter", DeviceID = "{CDCC59CD-8A52-4E59-A8A9-EDB73BC7F81A}", SystemCreationClassName = "CIM_NetworkPort", SystemName = "app-prod")
NetRoute                                : MSFT_NetRoute (InstanceID = "{8:8:9:55055;87:8;8;55;}")
SourceAddress                           : 10.30.1.4
NameResolutionSucceeded                  : True
BasicNameResolution                      : (Microsoft.DnsClient.Commands.DnsRecord_A)
LLMNRRecords                            : (Microsoft.DnsClient.Commands.DnsRecord_A)
DNSOverHttpsRecords                      : (Microsoft.DnsClient.Commands.DnsRecord_A)
AllNameResolutionResults                 : (Microsoft.DnsClient.Commands.DnsRecord_A)
IsAdmin                                  : False
NetworkIsolationContext                  : Internet
MatchingIPsecRules                       :

Name                                     : app-sits-prod-svc.azurewebsites.net
QueryType                                : CNAME
TTL                                       : 60
Section                                  : Answer
NameHost                                 : app-sits-prod-svc.privatelink.azurewebsites.net
app-sits-prod-svc.privatelink.azurewebsites.net  10      Answer      10.30.5.7

Name                                     : kv-sits-prod-app-svc.vault.azure.net
QueryType                                : CNAME
TTL                                       : 34
Section                                  : Answer
NameHost                                 : kv-sits-prod-app-svc.privatelink.vaultcore.azure.net
kv-sits-prod-app-svc.privatelink.vaultcore.azure.net  10      Answer      10.30.5.6

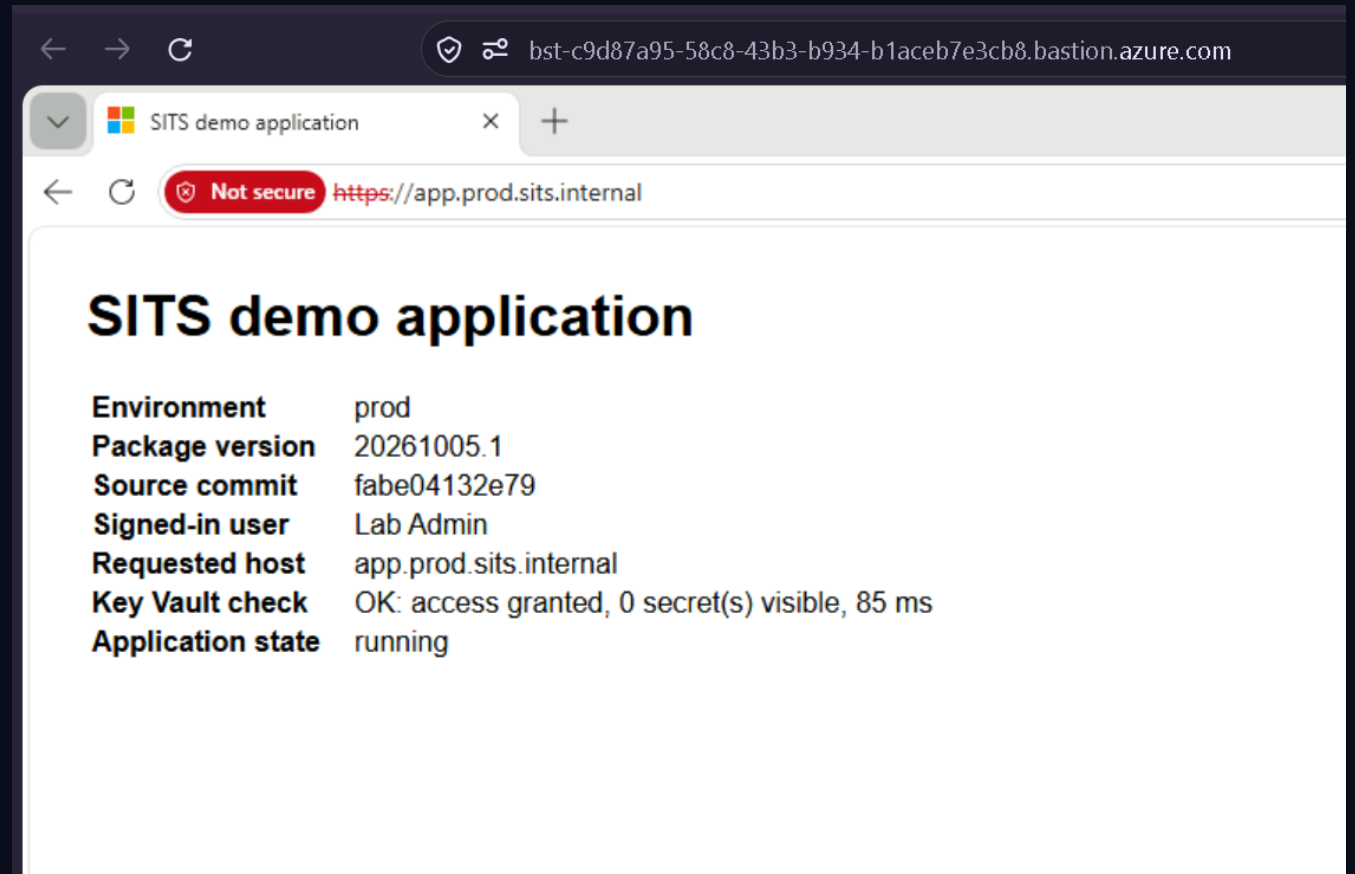
PS C:\Users\ProdAdmin> A_
```

Source: [docs/build-log/14-acceptance-checks-and-certificate-trust.md](#)

Application and identity

Configuration and observed results

The Python demo runs on Linux App Service with Entra Easy Auth. An assigned application group controls protected routes. The application managed identity reads the application vault, while a federation credential supports the sign-in configuration without an app-registration client secret. The /health route deliberately allows anonymous probing from network-authorized clients. Application access and Key Vault RBAC remain separate checks.



Source: [infra/environments/production/layer-2-application/appservice.tf](https://github.com/peterhudcovic/infra/environments/production/layer-2-application/appservice.tf)

Monitoring and diagnostics

Configuration and observed results

Gateway allLogs and AllMetrics stream to the production Log Analytics workspace using AzureDiagnostics tables. Dev and Test share the non-production workspace. Four custom alert rules per environment cover application health, unhealthy gateway backends, denied application-vault access and WAF matches. Production also has a Failure Anomalies smart detector. The Application Insights overview had no request telemetry in its selected one-hour window, so it does not prove active application instrumentation.

Home > agw-sits-prod-swc | Diagnostic settings

Diagnostic setting

Save Discard Delete Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#)

Diagnostic setting name: diag-log-sits-prod-swc

Logs

Category groups ⓘ

☒ allLogs

Categories

☒ Application Gateway Access Log

☒ Application Gateway Performance Log

☒ Application Gateway Firewall Log

Destination details

☒ Send to Log Analytics workspace

Subscription

sub-app-prod

Log Analytics workspace

log-sits-prod-swc (swedencentral)

Destination table ⓘ

Azure diagnostics Resource specific

☐ Archive to a storage account

☐ Stream to an event hub

☐ Send to partner solution

Metrics

☒ AllMetrics

Observed alerts

Configuration and observed results

The captured Azure Monitor view lists 16 fired instances across the selected subscriptions: 13 warnings and three informational alerts. These include repeated Key Vault access-denied results and WAF matches. WAF is in Detection, so a matched request does not establish prevention-mode blocking. The alert inventory and fired instances show configuration and observed events separately.

Home > Monitor

Monitor | Alerts

Microsoft

Search

Summarize alerts fired in last 24 hours

Help me setup a new alert rule

Catch me up on alerts fired this week

View as timeline (preview)

Create

Alert rules

Action groups

Alert processing rules

Prometheus rule groups

Change user response

Actions

Refresh

Alert rules

Create an autonomous observability agent: Automatically create and investigate issues from alerts, and correlate related alerts to reduce noise. [Learn more](#)

Create a new agent

Search

Subscription: 2 selected

Time range: Past 24 hours

Alert condition: Fired

Severity: all

Add filter

Total alerts

Critical

Error

Warning

Informational

Verbose

No grouping

Name ↑↓	Severity ↑↓	Affected resource ↑↓	Alert condition ↑↓	User response ↑↓	Fire time ↑↓
☐ alert-sits-prod-waf-request-matched	3 - Informational	log-sits-prod-swc	Fired	New	10/6/2026, 1:12 AM
☐ alert-sits-dev-waf-request-matched	3 - Informational	log-sits-dev-swc	Fired	New	10/6/2026, 1:09 AM
☐ alert-sits-test-waf-request-matched	3 - Informational	log-sits-dev-swc	Fired	New	10/6/2026, 1:08 AM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/6/2026, 12:59 AM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/6/2026, 12:49 AM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/5/2026, 11:34 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:27 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:22 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:22 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:21 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 11:06 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:46 PM
☐ alert-sits-prod-keyvault-access-denied	2 - Warning	log-sits-prod-swc	Fired	New	10/5/2026, 10:44 PM
☐ alert-sits-test-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:42 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:31 PM
☐ alert-sits-dev-keyvault-access-denied	2 - Warning	log-sits-dev-swc	Fired	New	10/5/2026, 10:26 PM

Source: docs/build-log/12-monitoring-alerts-and-infrastructure-pipelines.md

Limitations and next steps

Configuration and observed results

The current lab is deployed and demonstrated. Next work includes Terraform modules, a destroy-and-rebuild exercise, removal of temporary administrator state grants after approval, WAF Prevention after tuning, Conditional Access enforcement after report review, and a certificate from a trusted CA. Developer and tester infrastructure groups and app-user/app-denied demonstration accounts are not implemented. Foundation vault public exposure, shared monitoring dependencies and broad built-in roles remain visible design limits.

Sources and evidence

Repository snapshot 8def94695003

The main presentation follows the requested order: subscriptions, users and permissions, shared delivery, networking, Dev, Test, Prod, monitoring and next steps. The appendix preserves the collected screenshots and their English explanations.

Build records 12 and 14 record completed infrastructure pipeline checks, acceptance checks and certificate trust. The access matrix in build record 13 reflects an earlier snapshot; newer pipeline dependency grants are documented in build record 12 and the captured role assignments.

Application Insights overview is excluded because it displays a connection string. The original remains in local evidence. Identifiers, resource names and IP addresses are retained. Masked password fields contain no visible password value.

Project source: github.com/PeterHudcovic/azure-appservice-platform-lab